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System and method for transaction broadcast in a network on chip

Abstract

A broadcast adapter in a network-on-chip (NoC) is used for broadcasting transactions in the form of packets from an initiator to multiple targets and for receiving responses from the targets that are combined and sent to the initiator. The transactions originate from an initiator and are sent, using the NoC, to broadcast adapters using a special range of addresses. The broadcast adapters receive the transactions from the initiator. The broadcast adapters duplicate the transactions and send the duplicated transaction to multiple targets. The targets send a response, which is transported back by the NoC to the corresponding initiator.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) This application is a continuation of U.S. Pat. No. 12,038,866 (U.S. application Ser. No. 17/903,010 filed on Sep. 5, 2022) titled BROADCAST ADAPTERS IN A NETWORK-ON-CHIP which is a continuation of U.S. Pat. No. 11,436,185 (U.S. application Ser. No. 16/685,794 filed on Nov. 15, 2019) titled SYSTEM AND METHOD FOR TRANSACTION BROADCAST IN A NETWORK-ON-CHIP that issued on Sep. 6, 2022 to Syed 101 Ali SHAH et al. the entire disclosure of which is incorporated herein by reference.

TECHNICAL FIELD

(1) The present technology is in the field of system design and, more specifically, related to broadcasting transactions in a network-on-chip (NoC).

BACKGROUND

(2) System design of computer processors include multiprocessor systems. These multiprocessor systems have been implemented in systems-on-chips (SoCs) that communicate through network-on-chips (NoCs). The SoCs include instances of master (initiators) intellectual properties (IPs) and slave (targets) IPs. In some instances, one master sends a transaction or request to multiple slaves. The transactions are send using industry-standard protocols, such as ARM AMBA AXI, AHB or APB; or OCP-IP. The protocols have a strict request/response semantic, and typically are treated by a NoC as unicast: the master, connected to the NoC, sends a request to a slave, using an address to select the slave. The NoC decodes the address and transports the request from the master to the

slave. The slave handles the transaction and sends a response, which is transported back by the NoC to the master.

(3) The current known approach, when a master needs to send the same transaction or request to multiple slaves, is for the master to send all the requests sequentially. The master sends the transaction to the first slave, then to the second slave, then to the third slave and so on. For example, if a master wants to write the same data into 16 different slaves, the master sends 16 identical write transactions, in sequence, with one going to each slave. Thus, the time taken by the total operation—for sending 16 transactions—is 16 times the time of a single write transaction. This limits the rate at which an identical request can be sent to multiple slaves. The rate is limited by the rate at which the master can send sequential request to all the destinations, i.e. the slaves. Therefore, what is needed is a system and method that reduces the time taken to send multiple identical transactions from a master to multiple slaves.

SUMMARY OF THE INVENTION

(4) In accordance with various embodiments and aspects of the invention, systems and methods are provided to implement a new approach to sending a transaction from one master to multiple slaves. According to the various embodiments and aspects of the invention, a special range of addresses is used. The network-on-chip (NoC) broadcasts a transaction received at a special address, which is within the special range of addresses, to multiple destinations or slaves simultaneously instead of sending it to a single destination. One advantage is maximum efficiency of the operation that includes sending the same transaction to multiple destinations. Another advantage includes the ability to perform functions on a transaction prior to broadcasting the transaction.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 shows a network-on-chip (NoC) with a master and multiple slaves according to an embodiment of the invention.
- (2) FIG. 2 shows the NoC of FIG. 1 for a master broadcasting a write transaction to multiple slaves when the write transaction is sent to a first broadcast adapter (BA) according to an embodiment of the invention.
- (3) FIG. 3 shows the NoC of FIG. 2 when a write transaction is duplicated by the first BA and sent to other BAs according to an embodiment of the invention.
- (4) FIG. 4 shows the NoC of FIG. 3 when the other BAs send the write transaction to multiple slaves according to an embodiment of the invention.
- (5) FIG. 5 shows address ranges for broadcasting using the BA according to an embodiment of the invention.
- (6) FIG. 6 shows a BA for supporting broadcasting of a transaction according to an embodiment of the invention.
- (7) FIG. 7 shows a BA that includes multiple ingress ports for supporting multiple broadcast networks according to an embodiment of the invention.
- (8) FIG. 8 shows a BA that includes a transformation function module according to an embodiment of the invention.
- (9) FIG. 9 shows a BA that includes a buffer according to an embodiment of the invention.
- (10) FIG. 10 shows a flow process for broadcast a request from a master to multiple slaves using BAs in a special address ranges according to various aspects and embodiments of the invention.

DETAILED DESCRIPTION

(11) The following describes various examples of the present technology that illustrate various aspects and embodiments of the invention. Generally, examples can use the described aspects in any combination. All statements herein reciting principles, aspects, and embodiments as well as

specific examples thereof, are intended to encompass both structural and functional equivalents thereof. Additionally, it is intended that such equivalents include both currently known equivalents and equivalents developed in the future, i.e., any elements developed that perform the same function, regardless of structure.

(12) It is noted that, as used herein, the singular forms “a,” “an” and “the” include plural referents unless the context clearly dictates otherwise. Reference throughout this specification to “one embodiment,” “an embodiment,” “certain embodiment,” “various embodiments,” or similar language means that a particular aspect, feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment of the invention.

(13) As used herein, a “master” and a “initiator” refer to similar intellectual property (IP) modules or units and the terms are used interchangeably within the scope and embodiments of the invention. As used herein, a “slave” and a “target” refer to similar IP modules or units and the terms are used interchangeably within the scope and embodiments of the invention. As used herein, a transaction may be a request transaction or a response transaction. Examples of request transactions include write request and read request.

(14) Thus, appearances of the phrases “in one embodiment,” “in at least one embodiment,” “in an embodiment,” “in certain embodiments,” and similar language throughout this specification may, but do not necessarily, all refer to the same embodiment or similar embodiments. Furthermore, aspects and embodiments of the invention described herein are merely exemplary, and should not be construed as limiting of the scope or spirit of the invention as appreciated by those of ordinary skill in the art. The disclosed invention is effectively made or used in any embodiment that includes any novel aspect described herein. All statements herein reciting principles, aspects, and embodiments of the invention are intended to encompass both structural and functional equivalents thereof. It is intended that such equivalents include both currently known equivalents and equivalents developed in the future. Furthermore, to the extent that the terms “including”, “includes”, “having”, “has”, “with”, or variants thereof are used in either the detailed description and the claims, such terms are intended to be inclusive in a similar manner to the term “comprising.”

(15) Referring now to FIG. 1, a network-on-chip (NoC) **100** is shown in accordance with an embodiment of the invention. The NoC includes a master **102** in communication with a network interface unit (NI) **104**. The network interface units connected to slaves are used to convert the protocol used inside the NoC to the protocols used by the slaves. The NI **104** translates the incoming transactions, from the master **102**, to the protocol used inside the NoC **100** for transport. The NI **104** is in communication with a switch **106**. The switch **106** is in communication with a switch **108** and a switch **110**. The switch **110** is in communication with the switch **112**. The NoC **100** includes various pipeline elements in accordance with various embodiments of the invention, some of which are shown and some of which are not shown. The master **102** can communicate, through the NoC **100**, with slaves **130**, **132**, **134**, and **136**. The slave **130** communicates through a NI **120**. The slave **132** communicates through a NI **122**. The slave **134** communicates through a NI **124**. The slave **136** communicates through a NI **126**. In accordance with this embodiment of the invention, the master **102**, through the NI **104** inside the NoC **100**, communicates with four slaves **130-136** using four Nis **120-126**, respectively. It will be apparent that many other embodiments are contemplated with multiple masters and multiple slaves, even though only one master and four slaves are shown for clarity in this embodiments.

(16) In accordance with this embodiment of the invention, the NoC **100** also includes a broadcast adapter (BA) **142** in communication with the switch **112**, a BA **146** in communication with the switch **106**, and a BA **148** in communication with the switch **108**. The BAs, in accordance with the various aspects and embodiments of the invention, are connected to a request (transaction) network, as shown in FIG. 1 as well as the response (transaction) network side (the connections are shown in FIG. 6 in accordance with one embodiment of the invention).

(17) In accordance with the various aspects and embodiments of the invention, the BA **146** receives a packet (representing a request transaction or a request) on a request ingress port **150** (referred to also as an ingress port **150**). The ingress port **150** is on the request side of the transaction. There is a corresponding response ingress port on the response side of the transaction. The BA **146** duplicates the packet and sends the duplicates to each request egress port **152** and **158** (referred to also as an egress port **152** and **158**). According to the various aspects of the invention, the destination of each packet from each egress port **152** and **158** is set at the time of design.

(18) Considering the BA **146** as an example. A request packet of data (or request, which may also be referred to as a packet), which represents a transaction, arrives at the ingress port **150** of the BA **146**. In accordance with one aspect of the invention, the packet is duplicated and each duplicate packet is sent to each of the egress ports **152** and **158**. The egress port **158** sends one of the duplicated packets to the BA **148** through the switch **106** and then the switch **108**. The egress port **152** sends another one of the duplicate packets to the BA **142** through the switch **106** then the switch **110** and the switch **112**.

(19) In accordance with one embodiment of the invention, a packet arrives at an ingress port **178** of the BA **148**. The packet arriving at the ingress port **178** is duplicated. In accordance with an embodiment of the invention, the BA **148** includes an egress port **160** and an egress port **162**. The egress port **160** communicates with and sends packets to the slave (or target) **130** through the switch **108** and then using the NI **120**. Furthermore, the egress port **162** communicates with and sends packets through the switch **108** and then the NI **122** to the slave (or target) **132**.

(20) In accordance with one embodiment of the invention, any packet arriving at an ingress port **172** of the BA **142** is duplicated. In accordance with an embodiment of the invention, the BA **142** also includes two egress ports: an egress port **164** and an egress port **166**. The egress port **164** communicates with and sends packets to the slave **134** through the switch **112** and using the NI **124**. Additionally, the egress port **166** communicates with and sends packets through the switch **112** and the NI **126** to the slave **136**.

(21) Referring now to FIG. 2, FIG. 3, and FIG. 4, in accordance with embodiments of the invention, a write transaction **180** is originating from the master **102**. The master **102** is indicating that it is broadcasting the write transaction **180** to the slaves **130**, **132**, **134**, and **136** by sending the write transaction to the BA **146**. The master **102** sends the write transaction **180** to an address that is within the BA **146** range of addresses. The write transaction **180** arrives at the ingress port **150** of the BA **146**. The BA **146** duplicates the write transaction **180**. The BA **146** simultaneously sends the duplicated write transactions **180** through the egress ports **152** and **158**. One write transaction **180** arrives at the ingress port **178** (of BA **148**) through the switch **106** then the switch **108**. Another write transaction **180** arrives at the ingress port **172** (of BA **142**) through the switch **106**, the switch **110** and the switch **112**. The BA **148** and the BA **142**, each, duplicate the write transaction **180** arriving at their respective ingress ports. The duplicated write transaction **180** is sent from BA **148**, through the egress ports **160** and **162**, to the slaves **130** and **132**, respectively. The duplicated write transaction **180** is sent from BA **142**, through the egress ports **164** and **166**, to the slaves **134** and **136**, respectively. Thus, the master **102** is able to send a write transaction **180** to the slaves **130**, **132**, **134**, and **136** simultaneously.

(22) Referring now to FIG. 5, the NoC **100** includes a special range **500** of addresses that identify the BAs and standard address range **550** for each target or slave. As discussed, a BA duplicates a transaction that is received on its ingress port and sends the duplicated transaction to other elements, including other BAs, in the network using its egress port. When a master desires to initiate a broadcast operation and send a transaction to multiple slaves, then the master chooses an address from the address map that corresponds to a BA. The BA is like a target and has an address in the address map of the NoC. Thus, when a master sends a request with an address that matches an address for one of a BA, then the NoC will send the packet to that BA. The BA will then duplicate the transaction or request and send the duplicated transaction, in turn, to other

components (switches, pipelines, other BAs, or network interfaces) according to a pre-configured scheme.

(23) Referring now to FIG. 6, a BA **600** is shown with one request ingress ports and three request egress ports, and three response ingress ports and one response egress port, to handle a response coming from all slaves connected to the request egress ports in accordance with various embodiments of the invention. The BA **600**, on the response network portion, includes as many ingress ports as egress ports in the request direction: one response ingress port per request egress port. The BA **600** performs response aggregation and combines all the responses that correspond to one duplicate request packet, into a single response packet using a combination function. The combined response is sent back through the BA **600** response egress port.

(24) In accordance with some aspects of the invention, when the transaction is a write request, then one such combination function includes inspecting the write responses from the slaves for errors. If none of the incoming write responses contained an error, then the write responses are aggregated into a write response with no error. If any of the incoming write response contain an error, then the write responses are aggregated into a write response with an error. The aggregate write response is then sent back to where the request came from. The process is repeated until a write response is finally send to the master that made the initial write request.

(25) In accordance with some aspects of the invention, when the transaction from the master is a read request, then the read responses can be combined using a mathematical function such as addition, maximum, minimum and so on. The resulting combined read response is used as the read response to send back to where the request packet was coming from. The process is repeated until a read response is finally send to the master that made the initial read request.

(26) Referring now to FIG. 7, in accordance with one embodiment of the invention, a BA **700** is shown to support multiple different request type broadcast networks co-existing in a NoC. To support multiple broadcast networks, the BA **700** includes multiple request inputs or ingresses, one per broadcast network, to which the BA **700** is attached. In accordance with this embodiment of the invention, the BA **700** is connected to two broadcast networks. The NoC distinguishes between different broadcast networks by using a bit field in the packet header of a request transaction that is sent to the BA **700**. By setting the bit field appropriately, the desired broadcast network is selected from the multiple broadcast networks. The BA **700** sends duplicated packets on the selected broadcast network.

(27) In accordance with one embodiment of the invention, a BA includes the ability to select a particular set of request egress ports of the BA for a given packet that is received on the request ingress port. The packet received on the ingress port of the BA, is duplicated only onto the selected egress port. The selection of specific egress ports is implemented through dedicated selection bits in the header of the request transaction header (the packet header). The dedicated selection bits select the egress ports of the BA that a given packet shall be duplicated into for transmission. The egress ports of the BA, which are not selected, are marked as to be ignored for the response aggregation mechanism when the response transaction is received because no request was duplicated and sent through that specific egress port.

(28) Referring now to FIG. 8, in accordance with one embodiment of the invention, a BA **800** includes a transformation function for the payload of the transaction or packet. In one embodiment and according to one aspect of the invention, a transformation function includes conversion between different number formats, such as: integer to floating point or between different floating-point representation. Performing the transforming function on a packet payload in the BA **800** provides the advantage of doing the transformation function before the broadcast, wherein the write request is performed multiple times at multiple slaves. As such, the need for doing the transformation of the data at each slave is eliminated because each slave or target (destination) does not need to perform the transformation locally. For example, if an integer to floating point converter is implemented in the first BA (the BA **800**), then the master can send a write transaction

of an integer to the BA **800**. The BA **800** converts the integer into multiple writes requests of the corresponding floating-point representation before forwarding or sending the write request.

(29) Referring now to FIG. **9**, in accordance with one embodiment of the invention, a BA **900** includes a buffer **902**. The buffer **902** is a first in, first out (FIFO) buffer with one write pointer and one read pointer per egress port of the BA **900**. This buffer will permit independent progress of each egress port without having to implement one FIFO per egress port. The capability to make independent progress on each egress port permits freedom in implementation of complex broadcast networks while avoiding deadlocks. The buffer **902** behaves as follows: if one or more egress ports sees backpressure for a given packet FLIT, the FLIT is stored inside the buffer **902** in a FIFO order. Then the read pointer for the backpressured or blocked egress ports are set to that particular location and the write pointer of the buffer **902** advances. Previously blocked egress ports are reading their FLITs from the buffer **902** and each egress port has its independent read pointer inside the buffer **902**.

(30) Referring now to FIG. **10**, a process is shown for broadcasting to multiple slaves from one master in accordance with the various aspects and embodiments of the invention. The process begins, at step **1000**, by defining an address range, wherein the address range includes addresses for several BAs. At step **1100**, a master generates a request to send to a BA. At step **1200**, the master selects a BA and uses the address of the BA for the request. The request is received at the ingress port of the selected BA. At step **1300**, the BA adapter duplicates the request for transmission through the egress ports of the BA. At step **1400**, the BA sends duplicated requests to each slave connected to each of the BA's egress ports. As such, the master is able to broadcast a request simultaneously to several slaves using the address of the BA.

(31) Parallel processing can provide tremendous speedups. This is important for applications such as deep neural networks computations, which can require distribution of the same dataset to multiple nodes simultaneously. In accordance with some aspects of the invention, designers of neural network solutions with can take advantage of the BAs for implementing transaction completion in parallel or simultaneously. For example, various aspects and embodiments of the present invention can be implemented in the field of artificial intelligence computations and deep network accelerators. When implemented in hardware and software, such system can take full advantage of the parallelism of broadcasting using a NoC that includes BAs and run orders of magnitude faster.

(32) Certain methods according to the various aspects of the invention may be performed by instructions that are stored upon a non-transitory computer readable medium. The non-transitory computer readable medium stores code including instructions that, if executed by one or more computers, would cause the computer to perform steps of the method described herein. The non-transitory computer readable medium includes: a rotating magnetic disk, a rotating optical disk, a flash random access memory (RAM) chip, and other mechanically moving or solid-state storage media. Any type of computer-readable medium is appropriate for storing code comprising instructions according to various example.

(33) Certain examples have been described herein and it will be noted that different combinations of different components from different examples may be possible. Salient features are presented to better explain examples; however, it is clear that certain features may be added, modified and/or omitted without modifying the functional aspects of these examples as described.

(34) Various examples are methods that use the behavior of either or a combination of machines. Method examples are complete wherever in the world most constituent steps occur. For example and in accordance with the various aspects and embodiments of the invention, IP elements or units include: processors (e.g., CPUs or GPUs), random-access memory (RAM—e.g., off-chip dynamic RAM or DRAM), a network interface for wired or wireless connections such as ethernet, WiFi, 3G, 4G long-term evolution (LTE), 5G, and other wireless interface standard radios. The IP may also include various I/O interface devices, as needed for different peripheral devices such as touch

screen sensors, geolocation receivers, microphones, speakers, Bluetooth peripherals, and USB devices, such as keyboards and mice, among others. By executing instructions stored in RAM devices processors perform steps of methods as described herein.

(35) Some examples are one or more non-transitory computer readable media arranged to store such instructions for methods described herein. Whatever machine holds non-transitory computer readable media comprising any of the necessary code may implement an example. Some examples may be implemented as: physical devices such as semiconductor chips; hardware description language representations of the logical or functional behavior of such devices; and one or more non-transitory computer readable media arranged to store such hardware description language representations. Descriptions herein reciting principles, aspects, and embodiments encompass both structural and functional equivalents thereof. Elements described herein as coupled have an effectual relationship realizable by a direct connection or indirectly with one or more other intervening elements.

(36) Practitioners skilled in the art will recognize many modifications and variations. The modifications and variations include any relevant combination of the disclosed features. Descriptions herein reciting principles, aspects, and embodiments encompass both structural and functional equivalents thereof. Elements described herein as “coupled” or “communicatively coupled” have an effectual relationship realizable by a direct connection or indirect connection, which uses one or more other intervening elements. Embodiments described herein as “communicating” or “in communication with” another device, module, or elements include any form of communication or link and include an effectual relationship. For example, a communication link may be established using a wired connection, wireless protocols, near-filed protocols, or RFID.

(37) The scope of the invention, therefore, is not intended to be limited to the exemplary embodiments shown and described herein. Rather, the scope and spirit of present invention is embodied by the appended claims.

Claims

1. A network-on-chip (NoC) comprising: a plurality of network interfaces (NIs); a plurality of switches in communication with the NIs; and at least one broadcast adapter in communication with one or more switches of the plurality of switches, the broadcast adapter including: a request ingress port for receiving request packets; a response egress port for sending response packets; a plurality of request egress ports, wherein each request egress port is connected to a unique target and sends packets to the unique target; and a plurality of response ingress ports, wherein each response ingress port is connected to the unique target and receives packets from the unique target, wherein the broadcast adapter duplicates request packets that are received at the request ingress port and sends one duplicated packet to each request egress port of the plurality of request egress ports, and wherein the broadcast adapter checks each of a plurality of response packets, which are received at the plurality of response ingress ports, for errors when the duplicated request packets are write request transactions and combines the plurality of response packets to generate a combined packet that are received at the plurality of response ingress ports and sends the combined packet to an initiator using the response egress port.

2. A network-on-chip (NoC) comprising: a plurality of network interfaces (NIs); a plurality of switches in communication with the NIs; a plurality of edges connecting the plurality of switches to allow packets to travel from an initiator to a plurality of targets; and at least one broadcast adapter, which is at an address within a special address range and the address is selected by the initiator, in communication with one or more switches of the plurality of switches, the broadcast adapter including: a request ingress port for receiving request packets from the initiator; a response egress port for sending response packets to the initiator; a plurality of request egress ports, wherein each

request egress port is connected to a unique target selected from the plurality of targets such that the broadcast adapter sends a duplicated request packet, which is generated by the broadcast adapter, to the unique target; and a plurality of response ingress ports, wherein each response ingress port is connected to the unique target and the broadcast adapter receives a response packet from the unique target resulting in a plurality of response packets at the broadcaster adapter, wherein the broadcast adapter checks each of the plurality of response packets for errors when the duplicated request packet is a write transaction, wherein the broadcast adapter aggregates the plurality of response packets into an aggregated write response and sends the aggregated write response packet to the initiator using the response egress port, wherein the aggregated write response is an aggregated write response with an error when there is an error in any one of the plurality of response packets.

3. The NoC of claim 2 further comprising a transform module, wherein the transform module performs a function on at least one request packet of the request packets to generate a plurality of transformed duplicate request packets and each of the plurality of transformed duplicate request packets are sent through one request egress port of the plurality of request egress ports.

4. The NoC of claim 2 further comprising a buffer in communication with the request ingress port and the plurality of request egress ports, wherein the buffer stores data when there is backpressure on any given request egress port.

5. The NoC of claim 4 further comprising a control module in communication with the plurality of request egress ports, wherein the control module analyzes a control bit of at least one request packet and determines to which of the plurality of request egress ports the request packet is duplicated.

6. The NoC of claim 2, wherein there is one response ingress port for each of the plurality of request egress ports and wherein response packets received at the plurality of response ingress ports are combined to form a response packet at the response egress port.

7. A broadcaster of a network-on-chip (NoC), the broadcaster is selected by and accessed by an initiator using a special address range in order to allow the initiator generate a request that the broadcaster duplicates and sends to a plurality of targets, the broadcaster comprising: a request ingress port for receiving at least one request packet from the initiator; a response egress port for sending response packets to the initiator; a plurality of request egress ports, wherein each request egress port is connected to a unique target selected from the plurality of targets and sends a duplicated request packet, which is generated by the broadcaster, to the unique target; and a plurality of response ingress ports, wherein each response ingress port is connected to the unique target and the broadcaster receives a response packet from the unique target resulting in a plurality of response packets at the broadcaster, wherein the broadcaster checks each of the plurality of response packets for errors when the duplicated request packet is a write transaction, wherein the broadcaster aggregates the plurality of response packets into an aggregated write response and sends the aggregated write response to the initiator using the response egress port, and wherein the aggregated write response is an aggregated write response with an error when there is an error in any one of the plurality of response packets.

8. The broadcaster of claim 7, wherein the request includes a header having control bits that identify a set of request egress ports selected from the plurality of request egress ports for receiving the duplicate request thereby allowing the initiator to select which of the plurality of targets that are in communication with the broadcaster receive the duplicate request.

9. The broadcaster of claim 7 further comprising a buffer that includes: one write pointer; multiple read pointers; and one read pointer for each of the plurality of request egress ports.

10. The broadcaster of claim 7 further comprising: a plurality of response ingress ports in communication with the plurality of targets, such that there is one response ingress port for each of the plurality of request egress ports; and one response egress port in communication with the initiator, wherein response packets received at the plurality of response ingress ports are combined

to form a response packet that is sent to the initiator using the response egress port.

11. The broadcaster of claim 7, wherein the broadcaster includes a transformation function that converts a request packet payload of the at least one request packet before broadcasting when the at least one request packet is a write request thereby eliminating doing a transformation at each target.
